

TECH-ARCH-1252: Architecture Reference Document

Department: Architecture

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1. Document Scope and Executive Summary

This Architecture Reference Document (ARD) outlines the core policies, standard operating procedures, and guidelines for the A

This ARD applies to all employees involved in the design, development, testing, and deployment of software systems within [Co

2. Core Policies and Standard Operating Procedures

2.1 Architecture Principles

The following architecture principles serve as the foundation for all architectural design decisions:

- * Separation of Concerns (SoC): Divide system concerns into distinct layers or modules.
- * Single Responsibility Principle (SRP): Each module should have a single responsibility.
- * Don't Repeat Yourself (DRY): Minimize code duplication by using abstraction and inheritance.
- * KISS: Keep It Simple, Stupid. Favor simplicity over complexity.

2.2 Design Standards

The following design standards apply to all software products developed within [Company Name]:

- * Follow the Model-View-Controller (MVC) pattern for web applications.
- * Use the Entity-Attribute-Value (EAV) pattern for data modeling.
- * Implement security controls according to NIST 800-53 guidelines.
- * Adhere to the SOLID principles of object-oriented design.

2.3 Development Process

The following development process applies to all software projects:

1. Requirements gathering: Define project scope and functional requirements through collaboration with stakeholders.
2. Design phase: Create a comprehensive architecture design based on the requirements gathered.
3. Implementation phase: Develop the system according to the design specifications.
4. Testing phase: Perform unit testing, integration testing, and user acceptance testing (UAT).
5. Deployment phase: Deploy the system to production environments.

2.4 Change Management

The following change management process applies to all software products:

1. Request for Change (RFC): Submit a formal request for changes to the architecture or design.
2. Review and Approval: The Architecture Committee reviews and approves or rejects change requests.
3. Implementation: Implement approved changes according to the agreed-upon plan.

3. Compliance Monitoring, Penalties, and Operational Governance

3.1 Compliance Monitoring

The Architecture department is responsible for monitoring compliance with this ARD. This includes:

- * Regularly reviewing architecture designs and code reviews.
- * Conducting periodic audits of software products against these standards.
- * Providing training and awareness programs to ensure employees understand the policies and procedures.

3.2 Penalties

Failure to comply with this ARD may result in penalties, including but not limited to:

- * Verbal warnings
- * Written warnings
- * Suspension or termination of employment
- * Mandatory re-training or certification

3.3 Operational Governance

The Architecture department is responsible for the operational governance of software products. This includes:

- * Managing software product lifecycles from conception to retirement.
- * Ensuring compliance with industry standards and regulations.
- * Providing technical guidance and support to development teams.

By following this Architecture Reference Document, we can ensure the quality, security, and maintainability of our software products.